

Raghav Tiwari

Computer Science Student

NOIDA, India

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Portfolio: raghavtiwariportfolio.vercel.app

PROFESSIONAL SUMMARY

B.Tech Computer Science student focused on building AI-powered systems and data-driven applications. Experienced in developing LLM-based workflows, RAG pipelines, backend APIs, and interactive visualization tools using Python, FastAPI, and PostgreSQL. Interested in applied AI research, intelligent systems, and scalable software engineering.

TECHNICAL & CREATIVE SKILLS

Programming Languages: Python, C, C++, SQL, JavaScript, HTML, CSS

Frameworks & Libraries: Streamlit, Dash, React

Technologies & Concepts: Prompt Engineering, Retrieval-Augmented Generation (RAG), LLM Pipelines, Data Ingestion Pipelines, Generative AI, AI Agents, Data Analytics

Backend: REST APIs, API Integration, Async Processing, JSON, FastAPI

Databases: PostgreSQL, MySQL, VectorDB

Tools: VS Code, IntelliJ IDEA, Vercel, Git, GitHub, Antigravity

Methodologies & Professional Skills: Agile, SDLC, Research, Thinking

Design & Visual Communication: Canva, Photoshop, Typography, Branding, Layout Design

PROFESSIONAL EXPERIENCE

Data Management Specialist Intern

June 2025 – Jul 2025

Gaurs Group, Ghaziabad, India

- Streamlined data collection and organization by designing structured workflows for large-scale real estate datasets improving data accessibility and data integrity.
- Automated data validation and reporting processes using Python and Excel-based tools, reducing manual effort and turnaround time by approximately 40%.
- Collaborated with cross-functional teams to analyze operational data and support data-driven management decisions.
- Applied analytical methods to identify optimization opportunities in data handling and storage systems.
- Gained hands-on experience in data analysis, automation scripting, technical documentation, and structured data management.

PROJECTS

AquaMind | *FastAPI, PostgreSQL (pgvector), Google Gemini API, Dash*

Jul 2025 – Oct 2025

Github: github.com/RaghavTiwari31/AquaMind

- Developed a Retrieval-Augmented Generation (RAG) system for querying multi-year ARGO oceanographic datasets using natural language.
- Implemented backend APIs using FastAPI and PostgreSQL with pgvector for embedding-based semantic search.
- Integrated Google Gemini API to generate SQL queries, summarize results, and provide contextual insights from structured data.
- Built an interactive Dash-based web application for real-time chat, data visualization, and geospatial exploration of ocean parameters.

Built an MCP-based orchestration layer for persistent conversational memory and modular AI tool execution.

Github: github.com/RaghavTiwari31/Forensiq

- Built a graph-based financial forensics system that analyzes transaction networks to detect fraud patterns including circular fund routing, smurfing (fan-in/fan-out), and layered shell networks.
- Designed a detection pipeline that converts transaction CSV data into directed graphs and executes parallel graph-analysis pipelines to identify suspicious transaction clusters and fraud networks
- Implemented filtering mechanisms to reduce false positives from legitimate high-volume entities such as exchanges, merchants, and payroll systems.
- Developed an interactive D3.js visualization to explore suspicious transaction flows and network relationships.

PUBLICATIONS

- Raghav T., Adithya V., Dr. Nidhi S., Dr. Mudita N. (2025). **GreenShield – A Natural Language Processing Based Approach to Prevent Greenwashing and Attain Decarbonization**. In Springer Proceedings in Energy, Springer, Singapore. https://doi.org/10.1007/978-981-96-4492-6_6
 - Proposed an NLP-based framework for detecting corporate greenwashing in sustainability reports and advertisements.
- Raghav T., Adithya V., Prachi S., Dr. Deepika B. (2025). **How GenAI Is Replacing Humans: Myth or Real. Ethical Considerations in AI: Bias and Fairness in Generative Models**, Taylor & Francis. <https://doi.org/10.1201/9781003565703-3>
 - Explored the societal and ethical impact of Generative AI on automation, labor, bias, and fairness.

ACHIEVEMENTS

- **1st Place – IndustrySolve 2025 (Ideathon & Productathon), IIIT Delhi:** Secured 1st place in both Ideathon and Productathon rounds among 150+ teams, delivering a fully functional prototype that secured a ₹40,000 award.
- **Rift26-Physics Wallah Institute of Innovation: Secured 1st position** in our problem statement and **ranked 5th overall** among participating teams in a 24-hour national-level hackathon.
- **Vibecon, Emergent AI at IIT Delhi:** Built and presented an AI-driven solution during an invite-only 24-hour national hackathon.

EDUCATION

Guru Gobind Singh Indraprastha University
Bachelor of Technology in Computer Science
CGPA: 9.77 / 10

New Delhi, India
Aug 2023 – Aug 2027

CERTIFICATIONS

Graphic Design Masterclass – Udemy
Issued: May 2025